

Curcumin and neurodegenerative diseases

Over the last ten years curcumin has been reported to be effective against a wide variety of diseases and is characterized as having anti-carcinogenic, hepatoprotective, thrombosuppressive, cardioprotective, anti-arthritic, and anti-infectious properties. Recent studies performed in both vertebrate and invertebrate models have been conducted to determine whether curcumin was also neuroprotective. The efficacy of curcumin in several pre-clinical trials for neurodegenerative diseases has created considerable excitement mainly due to its lack of toxicity and low cost. This suggests that curcumin could be a worthy candidate for nutraceutical intervention. Since aging is a common risk factor for neurodegenerative diseases, it is possible that some compounds that target aging mechanisms could also prevent these kinds of diseases. One potential mechanism to explain several of the general health benefits associated with curcumin is that it may prevent aging-associated changes in cellular proteins that lead to protein insolubility and aggregation. This loss in protein homeostasis is associated with several age-related diseases. Recently, curcumin has been found to help maintain protein homeostasis and extend lifespan in the model invertebrate *Caenorhabditis elegans*. Here, we review the evidence from several animal models that curcumin improves healthspan by preventing or delaying the onset of various neurodegenerative diseases.